impl TradeoffCurve::to_profile

Michael Shoemate

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This proof resides in **"contrib"** because it has not completed the vetting process.

1 Hoare Triple

Precondition

Compiler-verified

- Receiver `self` of type `TradeoffCurve`

Caller-verified

The receiver is a valid tradeoff curve.

Pseudocode

```python
def tradeoffcurve_to_profile(curve: TradeoffCurve) -> PrivacyProfile:
    denom = CURVE_GRID_POINTS

    # beta_right[i] = beta((i + 1) / CURVE_GRID_POINTS)
    beta_right = []
    for i in range(1, CURVE_GRID_POINTS + 1):
        beta = curve.eval(i / denom)
        beta_exact = RBig.try_from(beta)
        beta_right.append(beta_exact)

    def profile(epsilon: float) -> float:
        if epsilon < 0.0:
            raise ValueError("epsilon must be a finite non-negative number")

        # Lower-bound exp(epsilon) so delta is rounded upward conservatively.
        epsilon = FBig.try_from(epsilon, rounding=Down)
        precision = max(epsilon.precision(), 10)
        epsilon = epsilon.with_precision(precision)
        exp_eps = RBig.try_from(epsilon.exp())  #

        delta = RBig(0)

        for i, beta in enumerate(beta_right):  #
            alpha = RBig(i) / RBig(denom)

            c1 = RBig(1) - exp_eps * alpha - beta
            c2 = RBig(1) - alpha - exp_eps * beta
            delta = max(delta, max(RBig(0), c1, c2))  #

```

```
30        result = inf_cast(delta)
31        return max(0.0, min(1.0, result))  # Clamp to [0, 1]
32
33    return PrivacyProfile(profile)
```

Postcondition

Theorem 1.1. Assume the receiver is a valid nonincreasing tradeoff curve $\beta : [0,1] \to [0,1]$. Then the pseudocode returns a valid conservative privacy profile $\delta(\epsilon)$ induced from β: for every $\epsilon \geq 0$, the returned value is no smaller than the exact (ϵ, δ) conversion of β.

Proof. Let $\beta : [0,1] \to [0,1]$ denote the input tradeoff curve. For any $\epsilon \geq 0$, the exact privacy profile induced by β is

$$\delta(\epsilon) = \sup_{\alpha \in [0,1]} \max\left(0,\, 1 - e^{\epsilon}\alpha - \beta(\alpha),\, 1 - \alpha - e^{\epsilon}\beta(\alpha)\right). \tag{1}$$

Let $d = \texttt{CURVE_GRID_POINTS}$. The pseudocode partitions $[0,1]$ into intervals

$$I_i = \left[\frac{i}{d}, \frac{i+1}{d}\right], \qquad i = 0, \ldots, d-1,$$

and stores the right-endpoint tradeoff values

$$\beta_i^{\text{right}} = \beta\left(\frac{i+1}{d}\right).$$

Let $\tilde{e}^{\epsilon}$ denote the computed value on line 19. By construction, line 19 rounds e^{ϵ} downward, so

$$\tilde{e}^{\epsilon} \leq e^{\epsilon}.$$

Fix any $i \in \{0, \ldots, d-1\}$ and any $\alpha \in I_i$. Since β is nonincreasing and $\alpha \leq \frac{i+1}{d}$, we have

$$\beta(\alpha) \geq \beta\left(\frac{i+1}{d}\right) = \beta_i^{\text{right}}.$$

Also, since $\alpha \geq \frac{i}{d}$, we have

$$\alpha \geq \frac{i}{d}.$$

Consider the first conversion term:

$$1 - \tilde{e}^{\epsilon}\alpha - \beta(\alpha).$$

Because $\tilde{e}^{\epsilon} \geq 0$, increasing α decreases this expression, and increasing $\beta(\alpha)$ also decreases it. Therefore,

$$1 - \tilde{e}^{\epsilon}\alpha - \beta(\alpha) \leq 1 - \tilde{e}^{\epsilon}\frac{i}{d} - \beta_i^{\text{right}}.$$

Similarly, for the second conversion term,

$$1 - \alpha - \tilde{e}^{\epsilon}\beta(\alpha) \leq 1 - \frac{i}{d} - \tilde{e}^{\epsilon}\beta_i^{\text{right}}.$$

Hence for every $\alpha \in I_i$,

$$\max\left(0,\, 1 - \tilde{e}^{\epsilon}\alpha - \beta(\alpha),\, 1 - \alpha - \tilde{e}^{\epsilon}\beta(\alpha)\right) \leq \max\left(0,\, 1 - \tilde{e}^{\epsilon}\frac{i}{d} - \beta_i^{\text{right}},\, 1 - \frac{i}{d} - \tilde{e}^{\epsilon}\beta_i^{\text{right}}\right).$$

Now compare exact arithmetic with the rounded value $\tilde{e}^{\epsilon}$. Because $\tilde{e}^{\epsilon} \leq e^{\epsilon}$, replacing e^{ϵ} by $\tilde{e}^{\epsilon}$ can only increase both conversion terms. Therefore, for every $\alpha \in I_i$,

$$\max\left(0,\, 1 - e^{\epsilon}\alpha - \beta(\alpha),\, 1 - \alpha - e^{\epsilon}\beta(\alpha)\right) \leq \max\left(0,\, 1 - \tilde{e}^{\epsilon}\frac{i}{d} - \beta_i^{\text{right}},\, 1 - \frac{i}{d} - \tilde{e}^{\epsilon}\beta_i^{\text{right}}\right).$$

The pseudocode computes exactly the maximum of the right-hand side over $i = 0, \ldots, d-1$ on line 28. Since the intervals I_i cover $[0,1]$, it follows that the returned value $\delta_{\text{approx}}(\epsilon)$ satisfies

$$\delta_{\text{approx}}(\epsilon) \geq \sup_{\alpha \in [0,1]} \max\left(0,\, 1 - e^{\epsilon}\alpha - \beta(\alpha),\, 1 - \alpha - e^{\epsilon}\beta(\alpha)\right) = \delta(\epsilon).$$

Finally, the cast on the return line rounds upward, and the clamp to $[0,1]$ preserves validity of the profile. Therefore the returned privacy profile is valid and conservative. $\square$